An autonomous career system for embodied creative research and development

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Autonomous agents are increasingly able to plan, monitor, and execute tool-mediated workflows, but a career objective is normally treated as motivation rather than as a measurable control problem. We report an autonomous career system that converts a target role, Walt Disney Imagineering Research and Development mechanical Imagineering in Glendale, into a logged state machine with role-fit dimensions, guardrails, review cycles, continuous manuscript updates, and public artifacts. The review loop uses GPT-5.5 when quota is available and a deterministic fallback when the model call fails; both paths use a whole-public-AO-Labs source graph including the CV, Sarrus, FluxCell, Relay, Relay Live, Progress, Curtis, Ocean, Talk, Yum, Lily, other public project surfaces, the WDI R&D profile, the active WDI listing, and Disney Research context. It can critique public sources autonomously but cannot send outreach, claim relationships, submit applications, or use private email/contact data without approval. At the current code lock on 15 May 2026, the production source registry also includes the PhD organization Google Doc text export, the public YouTube channel at `https://www.youtube.com/@nalalan`, Curtis practice endpoints, the Progress app itself, and additional AO Labs public surfaces. The live dashboard now uses a failure-cost evidence-ROI selector rather than the earlier weakest-signal rule alone. Candidate moves are scored by direct Disney-role alignment, bottleneck relief, current evidence created, compounding value, urgency, friction, and approval-gate penalty. The selected step is to lock one FluxCell experiment today by naming the motion target, hardware change, measurement, and first build date. The detailed analysis remains on the profile page, while Progress stores the PhD document as private server-side source state and exposes only source status, size, freshness, and derived current-step state publicly. The previous live fit signal was 74 of 100 with a credible-but-needs-signal confidence label; the same central constraint remains: Sarrus makes the active Glendale mechanical-design rung credible, while principal-scope ownership

and an active principal posting remain unverified. On 7 May 2026, the CV was revised to remove the redundant Sarrus subpage split and point the public record toward mechanical R&D, soft robotics, motion systems, and human-facing physical prototypes. On 15 May 2026, `https://progress.aolabs.io` was expanded from a general source ledger into the shared input layer for Imagineer, Curtis, Relay, the PhD planning document, YouTube practice media, papers, CV artifacts, and sibling AO Labs apps. The paper itself is part of the control loop. Any substantive change to a paper-backed AO Labs system must update the manuscript and PDF so the public artifact remains a cumulative record rather than a stale explainer. The contribution is therefore not a claim that the system has secured a job. It is a falsifiable protocol for making an otherwise ambiguous career-conversion process auditable, adaptive, continuously documented, and constrained by truthful public sources.

1 Introduction

Tool-using language-model systems can now reason, call external services, write artifacts, and update deployed software (*1*–*4*). These capabilities create an opportunity to treat long-horizon personal objectives as monitored systems rather than as episodic intentions, in the older spirit of autonomic systems and closed-loop discovery workflows (*5*, *6*). A career target is especially difficult because the desired outcome depends on evidence quality, technical fit, social trust, timing, applications, and human judgment. Without a state model, repeated effort can feel intense while remaining scientifically uninterpretable.

This paper describes an early deployment of an autonomous career-signal system aimed at WDI Research and Development mechanical Imagineering roles. The system uses the active WDI Research and Development Imagineer - Mechanical Design Engineer role as the live rung and the Principal R&D Imagineer - Mechanical Engineer profile as the north-star target (*7*). The positioning line is deliberately stable: mechanical PhD, soft robotics, creative prototyping, and AI-assisted tools for physical interaction systems.

The design borrows from controlled revenue experimentation: define a measurable state, choose the current bottleneck, constrain allowed actions, record interventions, and update a paper from real source material rather than from aspiration. The system is not allowed to invent credentials, inflate relationships, send spam, or apply on behalf of the operator. It can monitor state, update dashboards, maintain a paper, identify the next constraint, run a source-bounded AI review, and create or improve public artifacts that are grounded in existing work. The paper is not a one-time publication draft. It is a cumulative methods record, and it must change whenever the system changes in a way that affects evidence, policy, interface, metrics, sources, guardrails, or interpretation.

2 System architecture

The deployment consists of a public dashboard, a Railway-hosted backend, a JSON runtime state file mounted on a persistent volume, and scheduled or operator-triggered automation. The backend exposes health, operations-check, journal, weekly-paper, event, daily-cycle, and AI-review endpoints. As of 15 May 2026, the dashboard home page is intentionally a single action surface rather than an analysis surface. It reads the Imagineer backend and the Progress summary, selects one low-friction current step, and links directly to the source where the step begins. The profile page carries the longer state analysis. Progress carries the source inventory, scan history, private PhD document source state, YouTube source, Curtis practice state, Relay state, papers, CV, and sibling-app health.

The reviewer is a separate loop rather than a human-contact substitute. It builds a source bundle from the live state, the target listing, the WDI R&D profile, the paper PDF, the CV, Sarrus, FluxCell, Relay, Relay Live, Progress, Curtis, Ocean, Talk, Nerve, Duet, Violin, Yum, Lily, AO Labs home, Disney Research roster, and a Disney Research bipedal-character paper page. It also includes a compact identity profile summarizing the operator as a mechanical engineering PhD candidate building soft robotic materials, reconfigurable mechanisms, motion prototypes, human-facing physical systems, public project surfaces, research papers, and autonomous systems. The model returns source-bounded JSON: verdict, score, current constraint, signals, open limitations, candidate system work, and approval boundaries. If the model fails or times out, a deterministic fallback reviewer produces a conservative critique and records the fallback reason.

3 Decision policy

The policy now separates state scoring from next-step selection. State scoring still reports six role-fit dimensions: mechanical depth, creative prototyping, human-facing motion, principal signal, Glendale application profile, and the autonomous system itself. Each dimension reports the tracked signal, basis, and source-backed limitation that would move the score. The score basis is intentionally calibrated: a base prior from the state file, a bounded event bonus, a bounded portfolio bonus, a daily-cycle bonus, optional review-loop credit, and a readiness ceiling that names the unresolved blocker. A score of 100 is reserved for no known blocker in that lane, not for accumulated activity.

The next-step selector uses a failure-cost evidence-ROI policy. For each candidate action, the system assigns points for direct Disney-role alignment, relief of the current bottleneck, creation of current inspectable evidence, compounding value for future reviews, and urgency. It subtracts friction and approval-gate penalty. The selected action is the highest-scoring move that can start immediately without external outreach. This change matters because the weakest-signal rule alone can choose generic analysis work. The new policy favors actions that change the public record or the experiment state, especially when the current failure point is principal-scope ownership.

In the current lock, the weakest live signal is principal signal. The selected action is to lock one FluxCell experiment today: motion target, hardware change, measurement, and first build date. This is

Table 1. State variables at the current evidence lock. Values are reported from the live system and code state after the 15 May 2026 single-step and Progress-source update.

Variable	Recorded value
Target company and location	Walt Disney Imagineering R&D, Glendale, California
Live rung	WDI Research and Development Imagineer - Mechanical Design Engineer
North star	Principal R&D Imagineer - Mechanical Engineer
Fit signal	74 of 100 with credible-but-needs-signal confidence
Dominant bottleneck	Principal signal: visible ownership, technical direction, source depth, and approval-gated external validation
Counters	One daily cycle; fifteen AI-review records; zero outreach events; four portfolio anchors; five public artifacts; twenty-five journal entries
Active experiment	Autonomous career loop v0
Reviewer source graph	Imagineer reviewer sources now include Progress, the PhD organization document export, YouTube @nalalan, Curtis media and daily-record endpoints, Relay Live, Yum, Lily, papers, dashboards, and discovered AO Labs links
Human approval boundary	Applications, sensitive outreach, referrals, claims involving real people, and any private contact/email data
Progress source registry	Thirty-four configured Progress sources, including the PhD organization document, YouTube @nalalan, Curtis practice endpoints, Relay Live, Progress self-checks, papers, CV, and sibling AO Labs apps
Profile freshness	Public profile exposes the latest profile-state timestamp and source-count readout; the Imagineer home page shows only the current step
Paper continuity rule	Any substantive change to a paper-backed system updates the manuscript and PDF in the same work cycle

not another Sarrus-writing task. Sarrus is the completed technical anchor. The live failure point is that the public Disney case can become past-tense unless there is a measurable current R&D direction beyond the finished Sarrus paper.

This policy is intentionally conservative. It can make public artifacts, update the dashboard, refresh the manuscript, run autonomous critique, and record state. It cannot manufacture the social trust that a principal-level role requires. If an action requires a real person's attention or reputation, the system must stop at preparation and request approval before sending. The review loop therefore becomes a repeatable calibration instrument, not a license to automate human outreach.

4 Results

The current deployment is functional and has moved beyond the initial dashboard-only state. The dashboard and backend are live, the paper endpoint can publish a snapshot, the site exposes a PDF manuscript, and the AI-review path has produced live production critiques from the public source graph. The latest recorded review before the 15 May source expansion consumed 33 sources. The configured GPT-5.5 call returned an OpenAI quota error, so the deterministic fallback produced the current conservative review: score 74 of 100, credible mechanism depth, developing Disney motion signal, and thin principal-scope ownership signal. The source registry has since been expanded and the next scan is expected to record the new PhD document, YouTube, Curtis, Progress, and sibling-app sources.

The largest artifact change in this cycle was the Sarrus technical record. The system published geometry, compliant-link and flexural-joint path, pressure-driven expansion, module assembly, build state, force/stiffness figures, and motion examples. It then added an HTML-visible measurement table with pressure range, expansion ratio, estimated leg angle, output force, local stiffness, hysteresis status, planar dome height, double-dome resolution, and prototype limitation. The WDI R&D profile now uses this table as the dominant technical path while preserving the current data boundary: raw digitized force, stiffness, and hysteresis curves with uncertainty remain unresolved.

After the latest interface revision, the live site uses a compact operating surface instead of explanatory tiles or second-person coaching. The state surface now reports compact facts: Mechanical 86/100 with geometry, actuation, assembly, measurement, and motion; Principal 56/100 with ownership, technical direction, collaborators, and validation; System with 33 sources; profile freshness; and Boundary with applications, referrals, direct outreach, and person-facing claims. The type scale was reduced because the dashboard is an operating surface, not a poster. The outcome-signal cards use the same standard: direct signal, current source state, and system-owned operation. This is a functional requirement, not a tone preference.

The system-level fit signal is now calibrated to 74 of 100 after removing the earlier saturation behavior that let logged work push several lanes to 100. This should be interpreted as an internal readiness signal, not as an external hiring probability. The review result is deliberately separate: it scores the current public profile, not the probability of being hired. It confirms that the overall direction is credible for WDI R&D mechanical design while rejecting the idea that the current public profile is already principal-level. The profile now includes both a principal-scope boundary and an above-fold role-verification banner: the active target is the Glendale mechanical-design rung; the principal title remains a north-star profile until an active principal posting and principal-scope leadership work are independently verified.

The CV was revised in the same paper cycle because it is part of the reviewer source graph and the public application record. The selected-public-work section now treats Sarrus as one coherent project rather than splitting it into a separate subpage link. The top line and profile now point toward mechanical R&D, soft robotics, physical interaction, motion systems, and human-facing physical prototypes without naming the target directly. The Sarrus research and publication entries now use Sarrus-linkage soft-robotic cells and planar/cylindrical/locomotion assemblies instead of a generic topology frame. This keeps the CV aligned with the WDI R&D objective without reading like a named-company pitch or overstating principal-scope evidence.

The progress ledger adds a cross-application memory layer rather than replacing the Imagineer dashboard. Its first deployed version scanned eighteen public sources; the 15 May 2026 source registry now configures thirty-four sources. The added sources include the PhD organization Google Doc text export, YouTube @nalalan, Curtis ops, Curtis daily records, Curtis evidence progress, Relay Live, Progress self-checks, Wallguard, Spotify, Bus, and Violin. Progress stores the current PhD document text as a server-side scan source, exposes only status, size, and freshness on public routes, and returns the Imagineer current step as structured JSON. This creates the storage surface needed for long-term trend comparisons across papers, practice, portfolio state, public pages, planning logs, and outcome systems without publishing the private working log.

The Imagineer main page was reduced in the same cycle. The previous home page attempted to show

Table 2. Latest autonomous review result. The reviewer path is source-bounded over the public AO Labs graph; the latest production record used the deterministic fallback because the configured GPT-5.5 call returned an OpenAI quota error.

Field	Recorded value
Model	Deterministic fallback after OpenAI quota error; GPT-5.5 remains the configured model
Scope	Whole public AO Labs graph plus WDI role and Disney Research context
Source count	33 sources in the latest recorded review before the 15 May expansion; 34 configured Progress sources after the source-registry update
Review score	74 of 100
Verdict	Credible core, not inevitable yet
Top issue	Sarrus makes the mechanical case credible; principal-level ownership is not visible enough yet
Current system work	Single-step home page, profile analysis surface, Progress source ledger, PhD document ingestion, YouTube/Curtis source tracking, AO Labs Progress tile, continuous paper update

paper state, WDI state, lane scores, loop health, and signal analysis at once. That analysis now belongs on the profile page and in Progress. The home page now has one purpose: show the current small step, the blunt reason it matters, the expected time cost, the freshness timestamp, and a direct link to begin. The current step is to lock one FluxCell experiment today by naming the motion target, hardware change, measurement, and first build date. The visible reason is intentionally direct: if the FluxCell direction remains undefined, the Disney case stays past-tense, with a strong Sarrus paper but weak current R&D ownership.

The profile page now exposes its own update timestamp. This matters because the profile is not only a Disney-specific pitch; it is a compressed read of the public AO Labs record. New systems such as Curtis, unrelated project surfaces such as Yum, updates to Sarrus, paper endpoints, and progress-ledger snapshots are counted as context about the operator's range, persistence, taste, and systems-building pattern. They do not automatically increase role-fit scores. The scoring boundary remains explicit: only source material that strengthens mechanical R&D, physical interaction, prototype execution, systems ownership, or principal-scope responsibility should move the Imagineer readiness state.

5 Limitations and guardrails

Table 3. Role-fit dimensions after the current autonomous review cycle. Scores are internal operating metrics, not external hiring probabilities; the live dashboard exposes the tracked signal, basis, and unresolved signal for each row.

Dimension	Score	Current interpretation
Mechanical depth	86	Strong but not complete; Sarrus has a public technical record and measurement table, with raw digitized curves, uncertainty, and test conditions still pending.
Creative prototyping	82	Strongest signal; should remain attached to concrete physical demonstrations.
Human-facing motion	79	Credible signal, but still needs a concise guest-facing motion sequence.
Principal signal	56	Current bottleneck; visible ownership, technical direction, collaborators, and external validation remain thin.
Glendale profile	72	Credible for the active rung; CV framing is now aligned, while demo reel and role-specific narrative remain open.
Autonomous system	72	Backend, dashboard, durable state, reviewer, journal, and paper loop are live, but scheduled runs, evaluation history, and outcome tracking are early.

The system cannot guarantee an offer. A job decision depends on timing, hiring needs, competition, interview performance, portfolio interpretation, and institutional context. The useful claim is narrower: the system can make progress legible, choose the next bottleneck, prevent fake progress, and turn repeated effort into logged artifacts. It should be evaluated by public artifact velocity, critique quality, application readiness, interviews, and eventual conversion.

The guardrails are part of the result. The system forbids fabricated credentials, fake outreach, invented Disney relationships, automated applications, and claims that are not supported by public or logged sources. Private email/contact data is excluded from the reviewer unless explicitly approved. These constraints reduce short-term aggressiveness, but they preserve interpretability. If the system ever reports strong progress, that progress should be traceable to artifacts, logs, AI reviews, human approvals, applications, or outcomes.

6 Availability

The live dashboard is available at `https://imagineer.aolabs.io/`, with a fallback at `https://aolabs.io/imagineer/`. The live backend exposes `https://imagineer.aolabs.io/api/imagineer/ops-check`, `https://imagineer.aolabs.io/api/imagineer/weekly-paper`, and `https://imagineer.aolabs.io/api/imagineer/ai-review`. This manuscript is a public artifact and must be refreshed from logged sources in the same work cycle as substantive changes to the dashboard, backend, reviewer policy, source graph, public profile, Sarrus/portfolio evidence, metrics, or guardrails.

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